

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining**

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### COURSE OUTCOME COVERED

**CO4:** Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

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**QUESTIONS: 05**

**Total Marks:50**

Annexure A

### COMPARITIVE ANALYSIS

| <b>Criteria</b>  | Scope of Items (2) | Comparison Depth(2.5) | Use of Evidence (2) | Critical Thinking (2) | Clarity of Presentation (1) | <b>Total(10)</b> |
|------------------|--------------------|-----------------------|---------------------|-----------------------|-----------------------------|------------------|
| <b>Weightage</b> | 20%                | 25%                   | 20%                 | 20%                   | 10%                         | 100%             |
| <i>Q1</i>        |                    |                       |                     |                       |                             |                  |
| <i>Q2</i>        |                    |                       |                     |                       |                             |                  |
| <i>Q3</i>        |                    |                       |                     |                       |                             |                  |
| <i>Q4</i>        |                    |                       |                     |                       |                             |                  |
| <i>Q5</i>        |                    |                       |                     |                       |                             |                  |

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### **Code of Conduct:**

*I M. Farah Afrin /192511434 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Farah Afrin*  
**Signature of the student:**

## RUBRICS FOR CALCULATION:

|                         |                      | <b>Comparative analysis</b>                                                   |                                                 |                                            |                                          |
|-------------------------|----------------------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------|
| <b>Criteria</b>         | <b>Weightage (%)</b> | <b>Level 4 (Exemplary)</b>                                                    | <b>Level 3 (Proficient)</b>                     | <b>Level 2 (Developing)</b>                | <b>Level 1 (Beginning)</b>               |
| Scope of Items          | 20%                  | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant        |
| Comparison Depth        | 25%                  | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison            |
| Use of Evidence         | 20%                  | Supports comparison with accurate data, examples, or references               | Uses some supporting evidence with minor gaps   | Limited or weak supporting evidence        | No supporting evidence provided          |
| Critical Thinking       | 20%                  | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation; lacks depth          | No meaningful interpretation             |
| Clarity of Presentation | 15%                  | Well-organized, clear, and logically structured presentation                  | Generally clear with minor issues               | Somewhat unclear or poorly structured      | Disorganized and difficult to understand |

## Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes  
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

**Customer Income (₹) Spending Score**

|    |       |    |
|----|-------|----|
| C1 | 30000 | 60 |
| C2 | 60000 | 40 |
| C3 | 50000 | 70 |
| C4 | 28000 | 65 |
| C5 | 65000 | 35 |

**Question:**

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

**Document Word1 Word2 Word3**

|    |      |      |     |
|----|------|------|-----|
| D1 | 0.2  | 0.5  | 0.3 |
| D2 | 0.1  | 0.6  | 0.3 |
| D3 | 0.8  | 0.1  | 0.1 |
| D4 | 0.75 | 0.15 | 0.1 |
| D5 | 0.2  | 0.4  | 0.4 |

**Question:**

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

**Point Latitude Longitude**

|    |       |       |
|----|-------|-------|
| P1 | 13.08 | 80.27 |
|----|-------|-------|

|    |       |       |
|----|-------|-------|
| P2 | 13.10 | 80.25 |
| P3 | 12.97 | 80.22 |
| P4 | 13.05 | 80.30 |
| P5 | 13.00 | 80.20 |

**Question:**

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

**Product Sales (₹) Frequency**

|    |       |    |
|----|-------|----|
| P1 | 50000 | 20 |
| P2 | 30000 | 10 |
| P3 | 52000 | 22 |
| P4 | 25000 | 8  |
| P5 | 32000 | 12 |

**Question:**

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5 . Dataset: Mixed Attributes

**Item Price (₹) Category Rating**

|    |     |   |     |
|----|-----|---|-----|
| I1 | 500 | A | 4.5 |
| I2 | 300 | B | 3.8 |
| I3 | 550 | A | 4.7 |
| I4 | 250 | B | 3.5 |
| I5 | 520 | A | 4.6 |

**Question:**

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

## Q1 . Dataset: Customer Data

### Customer Income (₹) Spending Score

|    |       |    |
|----|-------|----|
| C1 | 30000 | 60 |
| C2 | 60000 | 40 |
| C3 | 50000 | 70 |
| C4 | 28000 | 65 |
| C5 | 65000 | 35 |

### Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'SimpleKMeans' algorithm is chosen, and the 'Cluster mode' is set to 'Use training set'. The 'Store clusters for visualization' checkbox is checked. The 'Clusterer output' pane displays the following information:

Within cluster sum of squared errors: 3.276367376753476

Initial starting points (random):

Cluster 0: C4,28000,65  
Cluster 1: C2,60000,40

Missing values globally replaced with mean/mode

Final cluster centroids:

| Attribute | Full Data | Cluster# 0 | Cluster# 1 |
|-----------|-----------|------------|------------|
|           | (5.0)     | (3.0)      | (2.0)      |

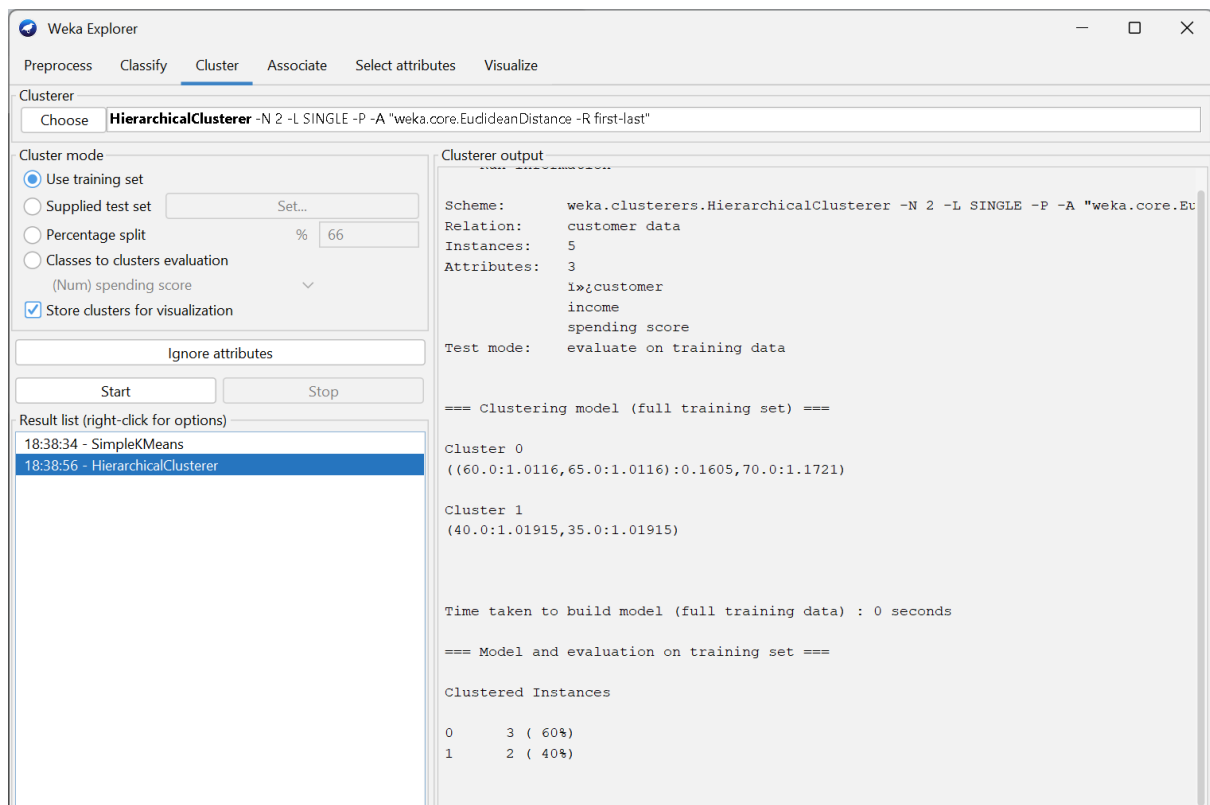
=====  
i>customer C1 C1 C2  
income 46600 36000 62500  
spending score 54 65 37.5

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 3 ( 60%) |
| 1 | 2 ( 40%) |



| Feature            | K-Means Clustering                           | Hierarchical Clustering                                    |
|--------------------|----------------------------------------------|------------------------------------------------------------|
| Method             | Divides data into a fixed number of clusters | Builds a hierarchy/tree of clusters                        |
| Number of clusters | Need to specify K in advance                 | Can decide the number of clusters from the hierarchy       |
| Speed              | Faster                                       | Slower                                                     |
| Scalability        | Better for large datasets                    | Less suitable for large datasets                           |
| Output             | Cluster assignments and centroids            | Dendrogram/tree structure                                  |
| Performance        | Efficient for large numerical datasets       | Useful for understanding relationships between data points |
| For this dataset   | Suitable and simple                          | Suitable, but more computationally expensive               |

## Q2. Dataset: Document Data

### Document Word1 Word2 Word3

|    |      |      |     |
|----|------|------|-----|
| D1 | 0.2  | 0.5  | 0.3 |
| D2 | 0.1  | 0.6  | 0.3 |
| D3 | 0.8  | 0.1  | 0.1 |
| D4 | 0.75 | 0.15 | 0.1 |
| D5 | 0.2  | 0.4  | 0.4 |

### Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.  
(BL4 – 1 Mark)

The screenshot shows the Weka Explorer interface with the SimpleKMeans clustering algorithm applied to the Document Data dataset. The 'Clusterer' tab is selected, and the 'SimpleKMeans' algorithm is chosen. The 'Cluster mode' section shows 'Use training set' selected. The 'Clusterer output' section displays the following information:

Clusterer output  
Cluster 0: D4,0.75,0.15,0.1  
Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:

| Attribute | Full Data | Cluster# | 0     | 1 |
|-----------|-----------|----------|-------|---|
|           | (5.0)     | (2.0)    | (3.0) |   |

=====  
i\document D1 D3 D1  
word1 0.41 0.775 0.1667  
word2 0.35 0.125 0.5  
word3 0.24 0.1 0.3333

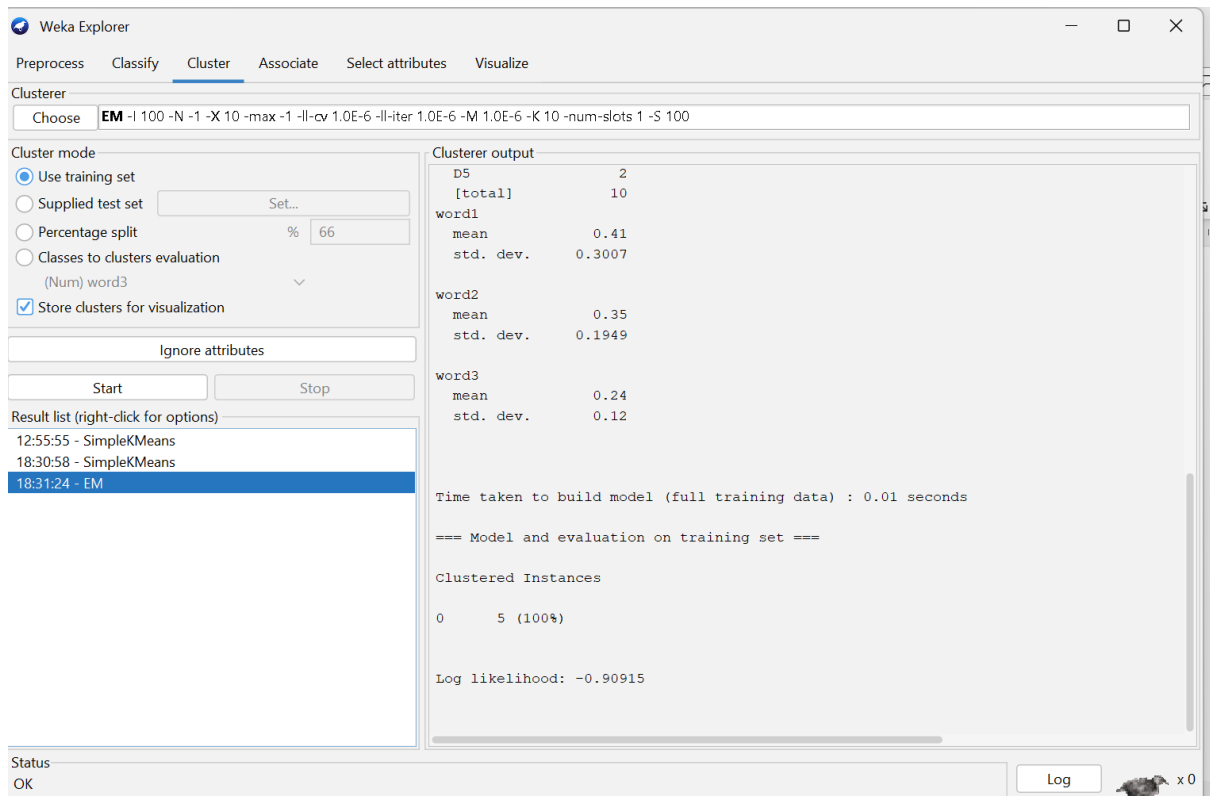
Time taken to build model (full training data) : 0.02 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 2 ( 40%) |
| 1 | 3 ( 60%) |

The 'Result list' on the left shows two entries: '12:55:55 - SimpleKMeans' and '18:30:58 - SimpleKMeans', with the latter selected. The 'Status' bar at the bottom indicates 'OK'.



### Q3. Dataset: Geographic Data

#### Point Latitude Longitude

P1 13.08 80.27

P2 13.10 80.25

P3 12.97 80.22

P4 13.05 80.30

P5 13.00 80.20

#### Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last" -I 500 -num-slots

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) longitude

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

18:39:58 - SimpleKMeans

Clusterer output

Sum of within cluster distances: 3.2482445759368774

Initial starting points (random):

Cluster 0: P4,13.05,80.3  
Cluster 1: P2,13.1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:

| Attribute | Full Data | Cluster# 0 | Cluster# 1 |
|-----------|-----------|------------|------------|
|           | (5.0)     | (2.0)      | (3.0)      |
| idpoint   | P1        | P3         | P1         |
| latitude  | 13.04     | 12.985     | 13.0767    |
| longitude | 80.248    | 80.21      | 80.2733    |

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 2 ( 40%) |
| 1 | 3 ( 60%) |

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.ManhattanDistance" -R first-last" -I 500 -num-slots

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) longitude

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

18:39:58 - SimpleKMeans

18:41:23 - SimpleKMeans

Clusterer output

Sum of within cluster distances: 5.61538461538462

Initial starting points (random):

Cluster 0: P4,13.05,80.3  
Cluster 1: P2,13.1,80.25

Missing values globally replaced with mean/mode

Final cluster centroids:

| Attribute | Full Data | Cluster# 0 | Cluster# 1 |
|-----------|-----------|------------|------------|
|           | (5.0)     | (1.0)      | (4.0)      |
| idpoint   | P1        | P4         | P1         |
| latitude  | 13.05     | 13.05      | 13.04      |
| longitude | 80.25     | 80.3       | 80.235     |

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 1 ( 20%) |
| 1 | 4 ( 80%) |

#### Q4. Dataset: Retail Products

##### Product Sales (₹) Frequency

|    |       |    |
|----|-------|----|
| P1 | 50000 | 20 |
| P2 | 30000 | 10 |
| P3 | 52000 | 22 |
| P4 | 25000 | 8  |
| P5 | 32000 | 12 |

##### Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'HierarchicalClusterer' is chosen as the clustering method. The 'Cluster mode' section shows 'Use training set' selected. The 'Clusterer output' pane displays the following information:

```
Scheme:      weka.clusterers.HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance"
Relation:    retail data
Instances:   5
Attributes:  3
              i>>product
              sales
              frequency
Test mode:   evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
(20.0:1.01286,22.0:1.01286)

Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

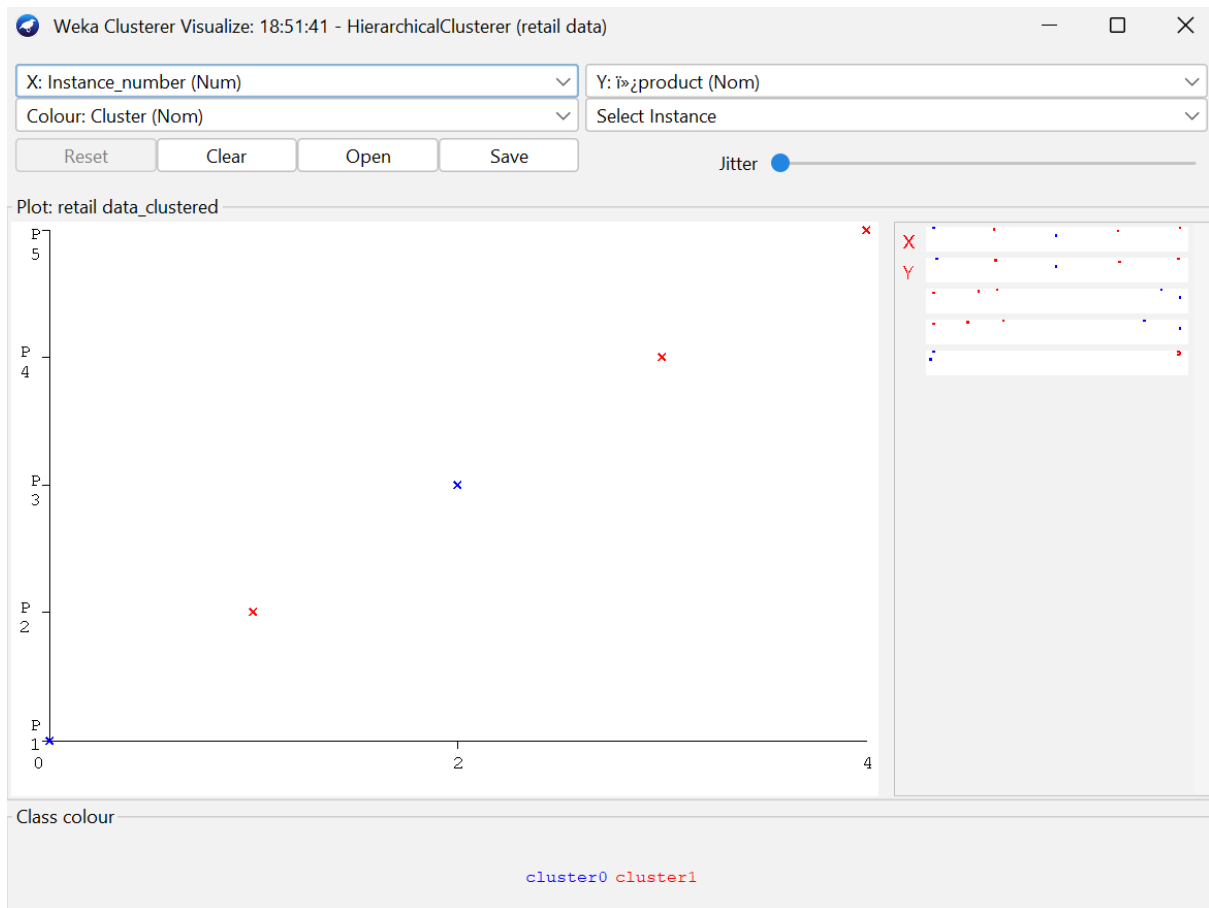
Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      2 ( 40%)
1      3 ( 60%)
```

The 'Result list' on the left shows three entries: '18:39:58 - SimpleKMeans', '18:41:23 - SimpleKMeans', and '18:51:41 - HierarchicalClusterer', with the latter being selected.



Q5 . Dataset: Mixed Attributes

**Item Price (₹) Category Rating**

|    |     |   |     |
|----|-----|---|-----|
| I1 | 500 | A | 4.5 |
| I2 | 300 | B | 3.8 |
| I3 | 550 | A | 4.7 |
| I4 | 250 | B | 3.5 |
| I5 | 520 | A | 4.6 |

**Question:**

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer  
Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last" -I 500 -num-slots

Cluster mode  
☒ Use training set  
☐ Supplied test set Set...  
☐ Percentage split % 66  
☐ Classes to clusters evaluation (Num) Rating  
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 18:39:58 - SimpleKMeans
- 18:41:23 - SimpleKMeans
- 18:51:41 - HierarchicalClusterer
- 18:54:18 - SimpleKMeans**

Clusterer output

Initial starting points (random):

Cluster 0: I4,250,B,3.5  
Cluster 1: I2,300,B,3.8

Missing values globally replaced with mean/mode

Final cluster centroids:

| Attribute | Cluster# |       |       |
|-----------|----------|-------|-------|
|           | 0        | 1     |       |
|           | (5.0)    | (2.0) | (3.0) |

=====

|          | I1   | I2   | I1       |
|----------|------|------|----------|
| Price    | 424  | 275  | 523.3333 |
| Category | A    | B    | A        |
| Rating   | 4.22 | 3.65 | 4.6      |

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 2 ( 40%) |
| 1 | 3 ( 60%) |

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer  
Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A

Cluster mode  
☒ Use training set  
☐ Supplied test set Set...  
☐ Percentage split % 66  
☐ Classes to clusters evaluation (Num) Rating  
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 18:39:58 - SimpleKMeans
- 18:41:23 - SimpleKMeans
- 18:51:41 - HierarchicalClusterer
- 18:54:18 - SimpleKMeans
- 18:54:35 - MakeDensityBasedClusterer**

Clusterer output

Normal Distribution. Mean = 275 StdDev = 25  
Attribute: Category  
Discrete Estimator. Counts = 1 3 (Total = 4)  
Attribute: Rating  
Normal Distribution. Mean = 3.65 StdDev = 0.15

Cluster: 1 Prior probability: 0.5714

Attribute: I1Item  
Discrete Estimator. Counts = 2 1 2 1 2 (Total = 8)  
Attribute: Price  
Normal Distribution. Mean = 523.3333 StdDev = 20.548  
Attribute: Category  
Discrete Estimator. Counts = 4 1 (Total = 5)  
Attribute: Rating  
Normal Distribution. Mean = 4.6 StdDev = 0.0816

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 2 ( 40%) |
| 1 | 3 ( 60%) |

Log likelihood: -5.93358